

OpenStreetMap: An Open Source Alternate to GoogleMaps

This article introduces OpenStreetMap, which is bound to interest students, engineers, trekkers, cartographers, government officials, law enforcement authorities and anyone who makes use of maps.

It doesn't matter whether you are a trekker, a cyclist, from a government planning agency, a traffic controller, a cop, a geocacher or a survivalist, there will most certainly be an instance where you would need a good map. People used to buy print maps or used the maps that came with travel guides. But those maps became outdated quickly, and it often took years to update changes that happened as a result of development, settlements and construction of infrastructure. After GoogleMaps was launched, the scenario changed. People rely on it extensively today. But there's a catch -- you're only allowed to view the maps in the browser or the Google Earth software. You are not allowed to share them in any way, to extrapolate data out of the maps, or even print parts of the map. That's a problem when you want to use the map to create a paper-based cycling map, or if you want to extract data for research. So, are there any online map providers who have *opened* up their maps? Enter OpenStreetMap (OSM for short).

OSM is a freely (as in freedom) editable world map. All the map (geographical) data contained in it is open-sourced (Open Data) and licensed under the Open Data Commons Open Database License (ODbL). The map cartography and tiles are licensed under a Creative Commons-Attribution-Sharealike licence. OSM is analogous to Google Maps. The map is rendered using Mapnik, an open source toolkit (LGPL).



Mapnik renders the 'Slippy Map' for OSM in the browser.

Mapnik can read from several formats like OSM's native XML, PostGIS, GeoJSON, ESRI Shape files, etc. Mapnik can be used independent of OSM, but how to do that is out of the scope of this article. Let's focus on OSM for now.

There are a number of reasons for using OSM:

1. Everything in OSM is free, as in freedom.
2. OSM is crowd-sourced.
3. It has very detailed map data.
4. Customised tiles are available.
5. It's very easy to embed and/or customise according to requirements.
6. You can access and edit OSM via API calls (REST-based, Yay!)
7. See <https://help.openstreetmap.org/>